Fiscal Year:	FY 2014	**Task Last Updated:**	FY 10/21/2013
PI Name:	Cromwell, Ronita Ph.D.		
Project Title:	Bed Rest Standard Measures		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	22
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	12/01/2004	**End Date:**	10/01/2015
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: Change in end date per PI (Ed., 3/12/2015)		
Key Personnel Changes/Previous PI:	September 2012: David Ware was removed from the protocol. October 2010: Buccello-Stout, Regina (Universities Space Research Association) Removed from protocol. Zanello, Susana (Universities Space Research Association) Added to protocol.		
COI Name (Institution):	Yarbough, Patrice (Universities Space Research Association) Zanello, Susana (Universities Space Research Association)		
Grant/Contract No.:	Directed Research		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The Flight Analogs Project Plan outlines a multi-phase approach for implementation of NASA-funded bed rest studies. This study is currently conducted at the NASA Flight Analogs Research Unit located at The University of Texas Medical Branch (UTMB) in Galveston, Texas. Although head-down bed rest has been used as a spaceflight analog to study changes in physiologic function associated with spaceflight for many years at many different institutions, standards for conducting such studies have not previously been established. Conditions of bed rest have been investigation or site specific. Most studies have had no direct comparisons of bed rest data with spaceflight data. As a consequence, systematic, multi-disciplinary comparisons across studies and comparisons of bed rest and actual spaceflight data have not been possible. To address these issues, the FAP has developed standard conditions for implementation of all NASA-funded bed rest studies and a set of standard measurements to be collected from all bed rest subjects. The standard measurements replicate, for the most part, measurements obtained on astronauts before and after long duration missions. Spaceflight and bed rest measurements are taken by the Johnson Space Center Laboratories and are used for three purposes: they make possible a direct comparison between responses to spaceflight and responses to head-down bed rest; they provide a characterization of human responses to long duration head-down bed rest; and they provide the ability to assess candidate countermeasures in a multi-disciplinary manner by evaluating the effects on the targeted systems as well as possible unexpected effects on non-targeted systems. The following measures are included in the standard measures complement. Nutritional status and immune function are assessed through blood, urine and saliva samples. Standard clinical laboratory assessment of blood and urine are used to monitor subjects' health status throughout the study. Dual Energy X-Ray Absorptiometry (DXA) in combination with bone biomarkers are used to assess changes in bone. Physical Fitness is assessed through measures of muscle strength, endurance, flexibility and maximum aerobic capacity. Posturography and monosynaptic T-reflex testing provides assessments of neurological function. The cardiovascular system is assessed through measures of orthostatic tolerance, blood volume, and cardiac function. These multidisciplinary measures are provided to investigators to enhance interpretation of system-specific results. By standardizing conditions and measures for all bed rest studies, NASA is able to compare results across bed rest studies in a multidisciplinary manner. The standard set of measures will be collected on as many bed rest subjects as is feasible to provide a multi-discipline evaluation of potential countermeasures. These measures will be integrated with the peer-reviewed science requirements on a non-interference basis.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	In FY2013, standard measures were collected for Campaign 11 (C11): Countermeasures and Functional Testing in Bed Rest. C11 is currently ongoing and is comprised of 6 integrated studies. One study identifies the key underlying physiological factors that contribute to changes in performance using a set of functional task tests that are representative of critical mission tasks for planetary operations. A second study examines the ability of an integrated resistive and aerobic training (iRAT) program to optimally mitigate the deleterious effects of long duration head-down tilt bed rest. A third study adds the use of a physiological dose of testosterone with the iRAT exercise program to enhance the effectiveness of exercise in preserving muscle and bone structure during long duration bed rest. The fourth study examines the relationship of smelling acuity and nasal congestion that result from six degree head-down tilt bed rest and how this impacts food acceptability and menu boredom. The aim of the fifth study is to validate an automated linguistic analysis to assess written journals and determine psychological state. The sixth study examines changes in neurocognitive function as related to neural structural alterations using structural and functional brain imaging techniques. The battery of standard measures was collected along with dependent measures for each study to assess possible changes in non-targeted physiological systems, and to compare counter measures across studies. Preliminary results of some of these measures are highlighted in this report. Subjects were randomly assigned to one of 3 main groups in a double blinded fashion: Non-exercising, Exercising, and Exercising with the testosterone supplementation. Because the study is currently ongoing, blinding of subjects in each group cannot be revealed. Therefore, subject data presented here are delineated into 3 groups without identification of each group. Bone Measures Biochemical markers of bone metabolism are monitored before, during, and after bed rest. A primary maker of bone resorption is N-telopeptide (NTX) as measured in the urine. Bone-specific alkaline phosphatase (BSAP) as measured in blood serum is a primary measure of bone formation. The percent changes in NTX values appear to increase after subjects enter bed rest. This was the case for all subject groups. For BSAP, one group appears to show greater bone formation than the others. Bone imaging studies using intelligent dual X-ray absorptiometry (iDXA) and quantitative computerized tomography (QCT) were also collected before and after bed rest. The iDXA bone mineral density (BMD) results for the lumbar spine showed a positive change in BMD for all groups ranging from 0.8 to 2.2 % over pre-bed rest values. Femoral neck BMD decreased from pre-bed rest values for all subject groups and ranged from -0.7 to -1.4 %. BMD in the trochanter and total hip decreased in 2 subject groups and increased in the third group. These values ranged from -2.1 to 1.9 % for the trochanter, and -1.6 to 0.28 % for the total hip. Finite element analyses of the QCT images was used to calculate bone strength for the hip under loading conditions simulating a lateral fall and single leg stance. Bone strength under lateral fall loading decreased after bed rest in all subject groups. Single leg stance loading decreased post bed rest for 2 of the 3 groups. One group showed no change in bone strength under stance loading after bed rest. Tilt Test Tolerance The tilt test assesses subjects' orthostatic tolerance to being tilted from supine to 80° for 15 minutes. Survival analyses are used to determine the probability that subjects could complete the 15 minute test without fainting. For 2 of the subject groups, the probability of completing the tilt test at the end of 70 days of bed rest was significantly less than before bed rest. In contrast, one of the subject groups did not show a significantly lower probability of completing the tilt test after 70 days of bed rest. Muscle Strength Isokinetic muscle strength was measured before and after bed rest. Results from knee extensor strength appear to show a decline in each subject group. Knee flexor strength tended to decline in one group while showing an increasing trend in the other groups. Trends in ankle extensor strength showed a decline for all groups, while ankle flexor strength remained similar across all groups.

Bibliography Type:	Description: (Last Updated: 05/16/2019)
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